

Roll No.

--	--	--	--	--	--	--	--

Paper Code: TCS-404/TIT-404

15/1/19

1.30-4.30

End Semester back Examination, 2018-19

B. Tech. (CS/IT) IV-Semester
Paper Name: Design and Analysis of Algorithms

Time: Three Hours

MM: 100

Note:

- i. This question paper contains five questions.
- ii. All questions are compulsory.
- iii. Instructions on how to attempt a question are mentioned against it.
- iv. Total marks assigned to each question are twenty.

1. Attempt any two questions of choice from a, b and c. (2*10=20)

(a) Explain all asymptotic Notations with help of example

(b) Derive the time complexities of the following function:

```
for(i=1;i<n;i++)
```

```
{for(j=n;j>1;j--)
```

```
{t=a;
```

```
a=b;
```

```
b=t;
```

```
}
```

```
}
```

(c) Solve $T(n) = 3T(n/3) + n$ using Recursive tree Method assume $T(1)=1$.

2. Attempt any two questions of choice from a, b and c. (2*10=20)

(a) Apply Merge Sort on following sequence to sort in non-decreasing order:

9 7 4 6 8 2 3 5

(b) Explain Fractional knapsack problem with help of example using greedy method.

(c) Write down algorithm for insertion sort and derive run time complexities.

3. Attempt any two questions of choice from a, b and c. (2*10=20)

(a) Explain the Subset-Sum Problem with the help of example. How it is solved using Backtracking.

(b) Describe the working of Branch and Bound technique. How Traveling Salesperson Problem is solved using this technique with the help of example.

(c) Explain 0-1 knapsack problem with example using dynamic programming.

4. Attempt any two questions of choice from a, b and c. (2*10=20)
- (a) **Design the Floyd-Warshall's Algorithm** for finding All-Pair Shortest Path. Explain with example.
 - (b) Design the algorithm for **BFS** and explain with example.
 - (c) Write down Prim's Algorithm. Explain with the help of example
5. Attempt any two questions of choice from a, b and c. (2*10=20)
- (a) Explain P, NP, NPC and NP-hard class of problem with suitable diagram.
 - (b) What is approximation Algorithms? Explain Vertex cover problem in this reference.
 - (c) What is reducibility? Explain with one example.